Micro Controller Based Voltage and Over-Current Protector

The device protects an appliance from grid voltage surges, sags and over-current.

It measures voltage and current, compares them with user-set limits, and opens a relay when a fault persists for 50 ms. After a trip the load stays off until the supply has been normal for a reconnect delay, which prevents relay chatter. Limits can be set locally with MENU/UP/DOWN buttons and a 16x2 I2C LCD, or from a computer over USB serial. Settings are stored in EEPROM.

- **Hardware:** Arduino Uno; voltage sensor (for simulation a POT is used) on A0; current sensor on A1 (for simulation a POT is used); relay on D7; LCD on A4/A5; buttons on D2-D4; buzzer on D6; green and red LEDs on D8/D9.

- **PC link:** 9600 baud text commands: GET, STATUS, SET OV/UV/OC/DLY, STREAM ON/OFF, RESET. STREAM sends DATA lines for logging or plotting.

Defaults: over-voltage 250 V, under-voltage 180 V, over-current 10 A, reconnect delay 5 s.

Test condition	Observed behaviour
Normal (about 220 V, low current)	Relay ON after reconnect delay, green LED lit, LCD shows LOAD ON
Voltage above 250 V	Relay trips, red LED and buzzer on, LCD shows TRIP:OVER-VOLT
Voltage below 180 V	Relay trips, LCD shows TRIP:UNDER-VOLT
Current above 10 A	Relay trips, LCD shows TRIP:OVER-CURR
Readings return to normal	Load reconnects automatically after the set delay
Settings changed (buttons or serial)	New limits take effect immediately and persist in EEPROM

How the idea is simulated in Wowki?

Wokwi has no mains source or AC sensor modules, so POT on A0 and A1 supply a steady 0-5 V that is scaled to 0-300 V and 0-20 A.

How the system can be scaled for real AC voltage measurement?

A ZMPT101B (voltage) and ACS712 (current) output a sine wave centred at 2.5 V.

The firmware must measure RMS instead of reading one value:

Sample every 200 µs (100 samples per 20 ms cycle) and collect whole cycles. Then accumulate the sum and the sum of squares. The mean is the DC offset, so RMS = sqrt(sum(x^2)/N - mean^2). Convert counts to amps or volts with the sensor sensitivity (ACS712-20A: 100 mV/A, 4.888 mV per ADC count), then calibrate voltage against a true-RMS meter. Track the peak value on every sample and trip at once on a short-circuit, without waiting for the RMS window.